

AutoTune: A Unified Benchmark for Highway Traffic Microsimulation Calibration

Cameron Hickert^{1,2,*}, Athena Wang³, Maryam Samaei⁴, Chengyuan Zhang⁵, Lijun Sun⁵, Yanbing Wang⁶, Mostafa Ameli^{4,7}, Cathy Wu^{1,2,8}

Abstract—While traffic microsimulation is an important and widespread tool for transportation research and engineering, its validity requires effective calibration. The community develops novel calibration algorithms with an assumption that progress is being made, but lacks a common benchmark to gauge these advancements. Algorithms’ opacity, implementation complexity, or proprietary data can hinder comparison, reproducibility, and collaborative research. To address this gap, this work introduces AutoTune, an open-source, large-scale microsimulation calibration benchmark built with data from the I-24 MOTION testbed. It enables robust assessment of calibration methods by providing microscopic and macroscopic ground-truth data, open-source implementations of key methods, and integration with popular car-following datasets. AutoTune supports calibrating diverse parameter categories to recreate traffic phenomena. A case study is presented to demonstrate its utility on a scenario of interest. AutoTune is available at <https://github.com/chickert/AutoTune>.

I. INTRODUCTION

Roadway transportation and infrastructure are important societal systems that remain largely underoptimized, resulting in significant issues such as traffic fatalities, time lost to traffic congestion, and considerable carbon emissions [1], [2]. Encouragingly, optimizing traffic flow via intelligent vehicles and infrastructure can lead to safer conditions and reduced emissions [3]. Traffic microsimulators (agent-based simulations at the level of individual vehicles) are an important tool for experimentation and evaluation in this domain.

Calibration aims to align the model parameters of a microscopic traffic simulator in a manner that accurately replicates observed traffic phenomena of interest [4].

A well-calibrated model is crucial for the reliability of subsequent analyses [5], ensuring that simulation outputs realistically mirror true behaviors. For example, well-calibrated microsimulation is essential for reliable evaluation of autonomous and connected vehicle control strategies and traffic management algorithms [6]–[9].

A core tenet of scientific advancement is objective comparison against a common yardstick. However, despite numerous calibration studies and methods – and similar benchmarking work for *macrosimulators* [10] – there is no widely accepted benchmark for *microsimulation* calibration that spans car-following, lane-changing, and origin-destination parameter types. This inhibits reproducibility and may disincentivize researchers from adopting new methods and settings, thereby siloing research efforts.

To address these gaps, this work makes **three contributions**. First, it proposes AutoTune, a standardized benchmark for highway microsimulation calibration that (i) leverages the I-24 Mobility Technology Interstate Observation Network (MOTION) traffic management and automated vehicle technology testbed, (ii) integrates with popular car-following model calibration datasets, and (iii) provides a suite of evaluation and analysis tools. Second, it selects and implements open-source versions of six popular or state-of-the-art microsimulation calibration approaches within the benchmark. Third, the work concludes with a case study to illustrate AutoTune’s value on a scenario of interest.

The benchmark’s name is derived from *automobiles* and parameter calibration, i.e., *tuning*. Furthermore, it is a reference to the benchmark’s focus on *automatic* calibration of traffic microsimulations. In one review of the level of effort required for freeway traffic microsimulation, model development, calibration, and data collection done manually on a single setting required between 180 and 4,500 hours [11]. This motivates the value of automatic methods. The benchmark is ‘unified’ in that it unites a range of parameter categories for calibration and a spectrum of objective functions, described further below. With these elements, AutoTune can enable more transparent, reproducible, and comparable research outcomes in this pivotal area of transportation science. See Figure 1 for a visualization of the road network.

¹MIT Laboratory for Information & Decision Systems, Cambridge, USA chickert@mit.edu, cathywu@mit.edu

²MIT Institute for Data, Systems, and Society, Cambridge, USA

³MIT Dept. of Electrical Engineering & Computer Science, Cambridge, USA athenaw@mit.edu

⁴GRETTIA, COSYS, Univ. Gustave Eiffel, Paris, France maryam-sadat.samaei@eleves.enpc.fr, mostafa.ameli@univ-eiffel.fr

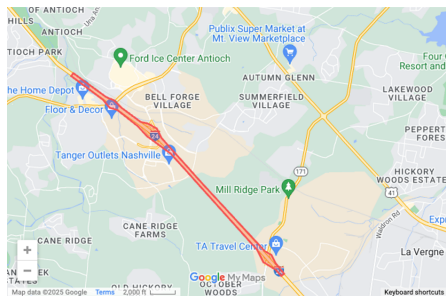
⁵Dept. of Civil Engineering, McGill University, Montreal, Canada enzozcy@gmail.com, lijun.sun@mcgill.ca

⁶School of Sustainable Engineering and the Built Environment, Arizona State University, Tempe, USA yanbing.wang@asu.edu

⁷Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, Berkeley, USA

⁸MIT Dept. of Civil & Env. Engineering, Cambridge, USA

* Corresponding author



(a) Google Maps view (rights reserved)



(b) Simulated version (with faded background)



(c) Closer view of simulated network

Fig. 1: Views of I-24 MOTION in Google Maps alongside AutoTune’s corresponding network constructed in the Simulation of Urban Mobility (SUMO) traffic microsimulator. Vehicles in the third panel are shown in yellow and ‘ghost detector’ locations are shown in gold with the true detector’s physical position shown in orange.

This work proceeds as follows: Section II describes the relevant literature and Section III details the problem. The benchmark’s design considerations and components are presented in Section IV, and Section V illustrates the benchmark’s utility with a case study. Section VI concludes.

II. RELATED WORK

Over the last 20 years, a growing body of research has focused on traffic microsimulation for planning, analysis, design, and future concepts [12]–[15]. This includes a range of research for intelligent vehicles: the work in [6] records 42 papers that leverage microsimulators for modeling connected and automated vehicles and that in [7] reviews 18 works using simulations to assess the operational and safety benefits of autonomous vehicles.

A parallel body of works studying calibration has more recently taken shape, particularly for car-following (CF) models and occasionally lane-changing (LC) models [5], [16]–[20]. Broadly speaking, these control *how* individual vehicles drive. However, it is known that (i) microsimulation performance is sensitive to other parameters, such as origin-destination (OD) pairs (*where* vehicles drive, often in aggregate), traffic heterogeneity, and simulator artifacts (such as time discretization) [21], and thus (ii) calibrating CF and LC parameters alone can fail to simulate traffic phenomena of interest [22]. Benchmarking work exists for OD flow calibration, but not for microsimulation [23], [24]. A benchmark analogous to AutoTune exists for macrosimulation calibration of continuum models [10], but this is insufficient for microscopic analysis, which is particularly important for intelligent vehicle applications, given that macro-simulators do not simulate dynamics at the level of individual vehicles [7]. Also, this benchmark is not open-source, inhibiting future comparison and development efforts.

We are only aware of two related open-source benchmarking efforts: [25] and [26]. However, the former

considers only CF parameters and relies on data from relatively few vehicle trajectories (39 total). The latter – although large-scale – is geared toward autonomous driving and thus comprises very short scenarios (9-20 seconds) that preempt OD specification and prevent the analysis of aggregate traffic phenomena of interest.

Thus, a gap remains for a long-term, large-scale, open-source traffic microsimulation benchmark that accommodates the broad range of parameters affecting a microsimulation.

A work of conceptual importance related to this topic is that in [4], which emphasizes how the broader optimization field’s ‘No Free Lunch’ result applies to the calibration of traffic simulations: namely, that no single method will dominate all scenarios. The work outlines several important practices that we incorporate into our benchmark, such as support for distributional error functions and stochasticity-aware assessment. Most importantly, this motivates the metric-agnostic problem formulation underlying our benchmark, detailed in the next section. Since no single objective can be universally optimal, the benchmark supports a range of objective functions. It also highlights the need for a unified approach to evaluating and sharing calibration approaches, so the community can identify the different scenarios in which generalization is warranted or not.

III. PROBLEM

Traffic microsimulation calibration consists of **selecting** and **setting** simulation parameters that produce realistic traffic. Because what constitutes ‘realistic’ traffic can include both individual trajectory-level as well as emergent macroscopic phenomena, the benchmark accommodates both trajectory and aggregated metrics.

A. Problem Formulation

1) *Highway Microsimulation Calibration:* Let $X = \{x_0, x_1, \dots, x_T\}$ represent a set of ground-truth vehicle

trajectory state data over a time horizon T . In practice, we rarely have direct access to X . Instead, we typically observe an aggregated (macroscopic) dataset of observations, denoted by $\mathcal{O} = h(X)$, where h is an aggregation function. This aggregation process, often involving averaging or summing over time and vehicles, mirrors how data is collected from sources like induction loops, radar detectors, or floating car data. Common examples of $\mathcal{O}$ include per-lane vehicle counts or average speeds over a specific time interval, and are available for select highways across several continents [27]–[29].

Let Θ represent the set of all possible traffic microsimulation parameters, which may be discrete or continuous. A θ -parameterized traffic dynamics model, denoted by F_θ , simulates the system state at time $t + 1$ according to the equation $x_{t+1} = F_\theta(x_t, u_t)$, where $\theta \in \Theta$ is the set of simulation parameters, x_t is the state at time t , and u_t represents the control inputs at that time. The sequence of simulated system states produced by F_θ is represented as $\tilde{X}_\theta = \{\tilde{x}_0, \tilde{x}_1, \dots, \tilde{x}_T\}$. From this set we can construct a simulation-generated dataset $\tilde{\mathcal{O}}_\theta = h(\tilde{X}_\theta)$.

We formulate the problem of **Highway Microsimulation Calibration** as identifying parameters according to

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} f(X, \tilde{X}_\theta, \mathcal{O}, \tilde{\mathcal{O}}_\theta). \quad (1)$$

Here, $f(\cdot)$ is an objective function that is a linear combination of both trajectory-level and macroscopic error terms:

$$f(X, \tilde{X}_\theta, \mathcal{O}, \tilde{\mathcal{O}}_\theta) = \sum_{t \in T} \sum_{g \in G} \left(\sum_{i=0}^m \alpha_i W_p(\hat{\mu}_{x_t}^{i,g}, \hat{\mu}_{\tilde{x}_t}^{i,g}) + \sum_{j=m+1}^n \alpha_j d_j(\mathcal{O}, \tilde{\mathcal{O}}_\theta) \right), \quad (2)$$

where g are spatial segments that collectively comprise the overall road network G , d_j are distance measures on desired aggregate metrics (such as detector count error, GEH statistic, etc.), and $\{\alpha_i, \dots, \alpha_n\}$ are n weight terms normalized such that $\sum_{i=0}^n \alpha_i = 1$, and are given and fixed. $W_p(\hat{\mu}_{x_t}^{i,g}, \hat{\mu}_{\tilde{x}_t}^{i,g})$ denotes the p -Wasserstein distance between the empirical distributions $\hat{\mu}_{x_t}^{i,g}$ and $\hat{\mu}_{\tilde{x}_t}^{i,g}$ of feature i extracted from x_t and $\tilde{x}_t$, respectively, on network segment g . Such features may include velocity, position, headway distance, or any other relevant characteristics of the system state at time t . The p -Wasserstein distance W_p then quantifies the dissimilarity between these feature distributions. $f(X, \tilde{X}_\theta, \mathcal{O}, \tilde{\mathcal{O}}_\theta)$ is assumed to exhibit a complex, non-convex topology in the high-dimensional parameter space.

Note that – aligned with [4] – this formulation subsumes common simulation objectives in the literature,

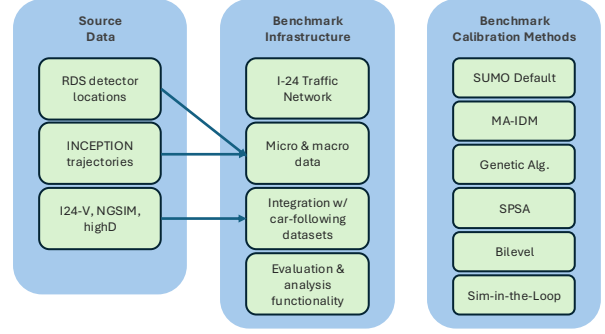


Fig. 2: Conceptual diagram of the benchmark and its source data.

including both trajectory-level and aggregated measures of performance and goodness-of-fit functions [25], [26]. For example, a traditional detector count root mean square error (RMSE) or speed distribution error can be recovered by setting the appropriate α_i or α_j terms to 0, and defining the remaining distance measure(s) as necessary. It also supports multi-objective optimization.

2) *Benchmarking*: Because of the complex topology of $f(\cdot)$, finding the optimal parameterization may be difficult without an exhaustive search, which is intractable given the large dimensionality and continuous elements in Θ . Calibration methods have varied approaches to navigating this and thus may exhibit different performance.

Let $\mathcal{C} = \{c_1, c_2, \dots, c_q\}$ be a set of q microsimulation calibration methods under consideration, where $f_k(\cdot)$ is the objective function score for the k th method. We consider the problem of **Benchmarking Highway Microsimulation Calibration** as evaluating these methods according to the objective function $f_k(\cdot)$, with lower scores indicating better performance.

IV. BENCHMARK

A. Why Highways?

AutoTune targets highways for multiple reasons. First, relative to minor roads, highways exhibit greater similarities across regions and countries [30]. Second, highways are important. Interstates host 26% of US vehicle miles traveled and 15% of US motor vehicle fatalities [1], [31]. Third, compared to minor road networks, highway networks exhibit simpler structure. This eases origin-destination and routing considerations, which we describe later. Fourth, aggregate data – including that collected via inductor loops, radar detection systems (RDS), and magnetic sensors – is relatively plentiful. For example, almost 40,000 detectors comprise California’s Caltrans Performance Measurement System (PeMS), which collects vehicle count and speed data in 30-second increments [32]. Finally, we now have access to over 40 hours of rich, trajectory-level data

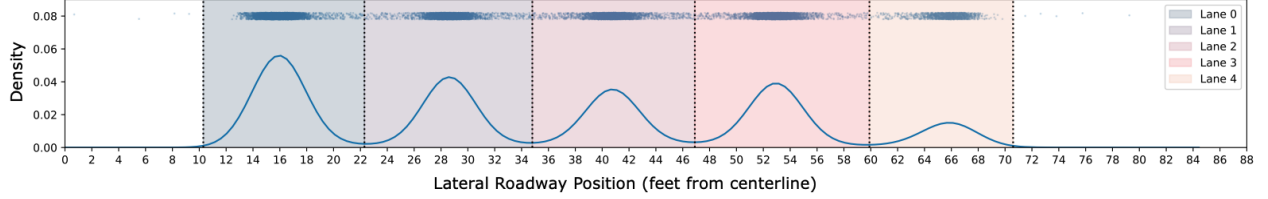


Fig. 3: Example of a kernel density estimate plot and the associated lane boundaries inferred from latitudinal position data at one westbound ‘ghost detector’. Blue dots mark the position of the underlying INCEPTION trajectories at this position along the roadway.

for traffic on a 6.75km stretch of highway via the I-24 MOTION project’s INCEPTION v1.0.0 data release, providing a high-fidelity dataset for assessing calibration methods [33]. As the INCEPTION releases continue, this resource will improve the quality and quantity of ground truth data for our benchmark.

B. Desiderata

AutoTune incorporates the desiderata outlined in the BetterBench checklist for benchmark developers [34]. While this is a checklist originally developed for artificial intelligence (AI) benchmarks, it draws on benchmarking practices in domains spanning bioinformatics, environmental quality, and transistor hardware. BetterBench outlines 47 best practices across multiple stages of the benchmark’s life cycle: design, implementation, documentation, and maintenance. We refer readers to [34] for the full list. 8 recommendations are not applicable – they suit AI benchmarks but not microsimulation calibration, such as API-based model evaluation and distinctions between local and cloud-based models. AutoTune satisfies 38 of the 39 that are applicable, ranging from the integration of domain literature to replicability scripts to code and data documentation. The unmet suggestion is the inclusion of a human performance level. However, because manual calibration can require hundreds of human hours per scenario, we have deliberately designed AutoTune to evaluate automated calibration methods that obviate the need for this labor.

One BetterBench requirement satisfied by AutoTune worth additional mention is open-source code and data availability. This is particularly important for our setting, given that simulation artifacts and other parameters or settings that may not be included in an academic publication or industry report can affect simulator performance. We thus have selected the Simulation of Urban Mobility (SUMO) traffic microsimulator.

In addition to BetterBench’s criteria, we follow other best practices that are specific to traffic microsimulation, such as the specification of calibrated parameters and ranges (see Appendix I), provision of necessary support files, and support for both trajectory-level (microscopic) as well as aggregate (macroscopic) traffic observations.

C. Components

An overview of the benchmark can be seen in Figure 2. It consists of (i) a SUMO replica of the I-24 MOTION traffic network, (ii) data and code necessary to process the I-24 INCEPTION data into calibration-usable microscopic and macroscopic data and the derived data itself, (iii) integration with popular car-following calibration datasets, (iv) implementations of relevant calibration methods, (v) an evaluation suite, and (vi) associated utility and analysis functionality. The processing in (iii) utilizes smoothing to mitigate known issues with computer vision-based vehicle trajectory identification methods [22], [35]. The suite in (v) provides native AutoTune support for a variety of error metrics – such as headway distribution divergences and detector- and segment-level speed discrepancies. Users can easily leverage others (conflicts, travel durations, etc.) that SUMO natively supports. It can run on a standard personal laptop, although more powerful, parallelized compute can accelerate calibration for methods that require numerous objective function evaluations. The code is available at <https://github.com/chickert/AutoTune> under an MIT license to aid in reproducibility. Also at this address is the necessary data and/or pointers to it.

1) *Inferring Lane Boundaries*: The AutoTune network includes the locations of 20 Tennessee Dept. of Transportation (TDoT) radar detection sensor (RDS) detectors positioned along the I-24 MOTION corridor. Unfortunately, TDoT RDS detector data is not available for the period covered by INCEPTION v1.0.0. Therefore, to extract aggregate data of the type common on highways from INCEPTION, we processed the INCEPTION trajectory data into ‘ghost detector’ aggregate speed and count data in 30-second increments at these positions (and include functionality for users to re-sample at different spatial and temporal frequencies, if desired).

One challenge that arose is the lack of lane information, which is a common feature in aggregated highway data [32]. We thus used a data-driven approach for lane inference via local minima searches in kernel density estimate plots of latitudinal vehicle positions at each detector location. See Figure 3. Once lanes are identified,

lane-level aggregated flow and speed data is derived from the INCEPTION trajectories.

2) *Data*: AutoTune provides processing code to directly interface with the I-24 MOTION INCEPTION v1.0.0 dataset [33]. It also includes highway detector-like measurements derived from the INCEPTION data, as described above. Combined, these offer a rich dataset of 47 hours of microscopic and macroscopic data from ~600,000 vehicles over 10 days across 6.75km of two-way interstate highway outside Nashville, TN. This is orders of magnitude larger in spatial and temporal scale than previous benchmarking efforts [19], [25]. It includes a variety of traffic behaviors of interest, such as free-flow traffic, rush-hour congestion, crashes, and police activity. AutoTune is a ‘living’ benchmark in that its utility will grow with future INCEPTION versions that contain more and higher-quality data.

We also provide integration functionality with three different datasets for car-following model calibration. Two of these are well-known (highD and NGSIM); we also include the I-24V dataset [36]. Given known issues with video-based trajectory identification, we use re-processed NGSIM data from [35] and post-process the I-24V data to mitigate detection errors. Current limitations include AutoTune’s restriction to a US highway setting, exclusion of urban networks (and related infrastructure like traffic signals, stop signs, etc.), and the current temporal, spatial, and quality data limitations of the INCEPTION v1.0.0 data.

D. Methods

Implementing calibration methods can be a time-consuming and sensitive task [11]. To address this, AutoTune includes open-source implementations of six calibration methods:

- **SUMO Default**: This uses SUMO’s default parameters as well as its built-in ‘flowrouter’ functionality to handle OD flow estimation. Often, researchers and practitioners use these defaults, so it is helpful to understand their performance and also presents a useful baseline [6], [7].
- **Memory-Augmented Intelligent Driver Model (MA-IDM)**: Bayesian methods are popular for CF model calibration. MA-IDM is a recent method that updates Bayesian calibration to address the reality that vehicle trajectories do not conform to i.i.d. assumptions [37]. We refer the reader to the original work for more details. This method may leverage smoothed leader-follower pairs extracted from the I-24V dataset, NGSIM, or highD.
- **Genetic algorithm (GA)**: This method is the most popular method identified in our literature review [5], [19], [25]. In our implementations, GA

and SPSA calibrate the widest range of parameters, described further below.

- **Simultaneous perturbation stochastic approximation (SPSA)**: This was the second most popular calibration method in our review [4], [38]. Both GA and SPSA are gradient-free approaches, making them useful in a variety of optimization settings.
- **Bilevel**: Adapted from [38], this approach optimizes two distinct objective functions independently, one at a level with coarse data and one with fine-grained data. We refer the reader to the original work for more details. It is included for its suitability for datasets commonly available for road networks, such as inductor loop detector data. Also, this method and the one below were recently applied to the I-24 MOTION setting and thus represent a characterization of the state-of-the-art for this area.
- **Sim-in-the-Loop (SL)**: Adapted from [39], this is a straightforward simulation-in-the-loop calibration framework that uses the Optuna optimization framework and is targeted at CF and LC parameters [40]. See the original work for more details. Because the original work requires OD demand to be predefined, one is supplied via a heuristic method building flows from the aggregated detector data.

The SUMO Default calibrates OD flows, albeit relatively naively. MA-IDM calibrates CF parameters. SL calibrates CF and LC parameters. The Bilevel method calibrates select LC and OD parameters. GA and SPSA calibrate the most parameters, including OD flows, CF and LC models, traffic heterogeneity, and simulator artifacts. For computational tractability, these use a block optimization approach. Where parameters are unspecified in these methods, SUMO Default values are used. See Appendix I for more details on parameters and categories.

V. CASE STUDY

1) *Scenario*: To demonstrate AutoTune’s utility, we present a case study on a scenario encompassing east-bound and westbound traffic between milemarker 58.8 and 62.8 on I-24 from 6am to 10am on Nov. 28th, Nov. 30th, and Dec. 2nd (Monday, Wednesday, Friday, respectively) in 2022. These are selected to include a range of phenomena of interest. The 30th represents a standard morning of rush-hour traffic. On the 28th, police pulled over a vehicle on the westbound corridor during rush hour. On Dec. 2nd, two vehicles crashed. We are interested in assessing how the calibrated simulations compare to the known data for these days – a common use case for traffic simulations. Thus, here we do not use a separate validation set.

As our objective function for comparing the methods, we use lane-level counts at each detector every

Difference Between Generated and Measured
Traffic Counts (Westbound, 6:00 AM - 7:00 AM)

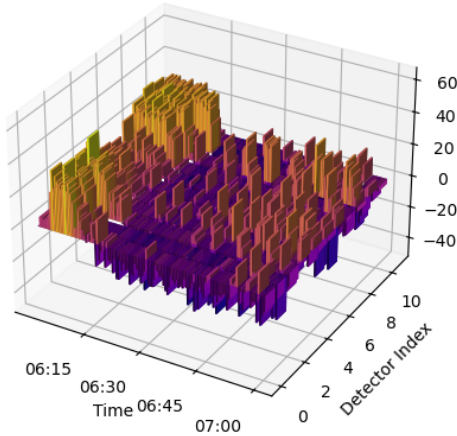


Fig. 4: Snapshot of the discrepancy between simulated and INCEPTION data for the SUMO Default method at the simulation’s start. Higher detections in simulation than the original data are shaded yellow, while lower are shaded purple.

30 seconds as our measure of performance, and select root mean square error (RMSE) as our goodness of fit function. This is a commonly used objective function on related benchmarks and calibration works and represents a throughput assessment task of potential use for academic, government, and industry users [5], [6], [10], [25], [41]. While this was chosen for the current case study, AutoTune supports a variety of other microscopic and macroscopic metrics, as described in Section IV-C. The GA and SPSSA methods use repeated evaluations of this objective function in their optimization process, while the SUMO Default and MA-IDM do not iteratively optimize.

To more faithfully implement the Bilevel and Sim-in-the-Loop methods as described in their original works, these algorithms use their original objective functions. This also enables comparison along that dimension, relevant to the No Free Lunch theorem.

After experimenting with the I-24V, NGSIM, highD datasets, the processed I-24V trajectories were found to enable the best MA-IDM performance on this scenario, so it was used. The GA and SL methods were each allowed a budget of 5,000 simulations during their calibration process, parallelized across up to 80 separate CPUs. The SPSSA method, which does not support parallelization, was allowed a budget of 3,000 simulations. The block optimization for the GA and SPSSA methods grouped parameters according to their parameter category, with an equal number of objective function evaluations allocated to each block. The order was as described in Appendix I. Five iterations through

TABLE I: RMSE computed from lane-level detector measurements taken at 30-sec. increments for each method on the various days. The standard result is for Nov. 30th. Police activity occurred on Nov. 28th. Crashes occurred on Dec. 2nd. **Bold** values indicate the best (lowest) RMSE on each day.

Method	Standard	Police	Crash	Average
SUMO Default	4.98	4.76	5.06	4.93
MA-IDM	6.10	6.39	6.81	6.43
GA	3.70	3.57	3.85	3.71
SPSSA	3.76	3.65	3.90	3.77
Bilevel	3.89	3.87	3.80	3.85
SL	6.70	7.25	6.90	6.95

the entire block cycle were used, with the budget of simulations evenly divided among each iteration.

2) *Results & Discussion:* The results are shown in Table I. Averages are included to account for stochasticity in the methods. Overall, the methods that calibrate a wider variety of parameters (GA, SPSSA, Bilevel) outperform those that do not (MA-IDM, SL, SUMO Default). MA-IDM’s under-performance relative to the SUMO Default method suggests that calibrating CF parameters alone can actually hurt performance, even when using trajectories drawn from the same roadway (albeit, at a different time). This may be a result of overfitting to different conditions or may indicate mismatch with OD flows. This supports AutoTune’s emphasis on holistic calibration for microsimulation.

We hypothesize that the SL method’s relatively poor performance is a result of the heuristic OD flow mechanism needed to adapt it to AutoTune, rather than the optimization process itself. This is because, while it steadily improved during optimization, it began with a simulation that produced higher errors than the others. Future efforts may find that adapting flow calibration from other methods can address this.

The discrepancy in performance of the Bilevel and SL methods – each of which used a different objective function than the other approaches – suggests a direction for future work investigating the practical impact of the No Free Lunch theorem in settings of interest. In particular, the Bilevel approach’s relatively strong performance suggests generalization across objectives may be feasible in some settings. Notably, the Bilevel method demonstrates the most robust performance across the three days, maintaining low error rates despite variations in traffic conditions.

AutoTune provides visualization tools to analyze results, such as count plots, time-space diagrams, and metric distribution visualization. One example is shown in Figure 4, which shows the discrepancy between the simulation and the SUMO Default method for the first hour of simulated time in the westbound direction. It can

be seen that the simulation initially introduces too many vehicles, followed by a period of too few, suggesting the simulator’s traffic demand profile could benefit from further calibration. Tools such as this can aid in assessing and improving methods’ performance.

VI. CONCLUSION

This work introduced AutoTune, an open-source, large-scale benchmark for highway traffic microsimulation that leverages real-world data from the I-24 MOTION testbed. Implemented on this benchmark are open-source adaptations of six popular and state-of-the-art calibration methods, enabling direct, reproducible comparison, as demonstrated with a case study utilizing traffic across 12 hours on three separate days with varied traffic conditions. Future work can (i) leverage AutoTune to more rigorously understand the practical implications of the No Free Lunch theorem for settings of interest, (ii) assess methods’ robustness across data types, objective functions, and scenarios, and (iii) further analyze the relative importance of various parameter types. AutoTune can thus enable the development and evaluation of more efficient and accurate microsimulation calibration.

APPENDIX I CALIBRATION PARAMETERS & CATEGORIES

Table II details the candidate parameters natively exposed by AutoTune and their default ranges. While this list constitutes the primary search space, users may extend the benchmark to include additional simulation parameters as desired. Note that specific methods often optimize a subset of these variables (or method-specific alternates) as described in Section IV-D. Unless a method specifies otherwise, this benchmark primarily utilizes the Intelligent Driver Model (IDM) for car-following and the Erdmann Model for lane-changing due to their prevalence [42]. Users may substitute alternative car-following or lane-changing models. For the methods using block optimization (GA and SPSA), the parameter category ordering in the case study occurred according to the order listed in Table II, and all parameters considered were as described in the table. Users may change this order as desired to experiment with different combinations.

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REFERENCES

- [1] A. M. Welki and T. J. Zlatoper, “The impact of highway safety regulation enforcement activities on motor vehicle fatalities,” *Transportation research part E: logistics and transportation review*, vol. 43, no. 2, pp. 208–217, 2007.
- [2] V. Jayawardana, B. Freydt, A. Qu, C. Hickert, E. Sanchez, C. Tang, M. Taylor, B. Leonard, and C. Wu, “Mitigating metropolitan carbon emissions with dynamic eco-driving at scale,” *arXiv preprint arXiv:2408.05609*, 2024.
- [3] A. R. Kreidieh, C. Wu, and A. M. Bayen, “Dissipating stop-and-go waves in closed and open networks via deep reinforcement learning,” in *2018 21st international conference on intelligent transportation systems (itsc)*. IEEE, 2018, pp. 1475–1480.
- [4] B. Ciuffo and V. Punzo, ““no free lunch” theorems applied to the calibration of traffic simulation models,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 15, no. 2, pp. 553–562, 2013.
- [5] V. Punzo, B. Ciuffo, and M. Montanino, “Can results of car-following model calibration based on trajectory data be trusted?” *Transportation research record*, vol. 2315, no. 1, pp. 11–24, 2012.
- [6] N. Raju and H. Farah, “Evolution of traffic microsimulation and its use for modeling connected and automated vehicles,” *Journal of Advanced Transportation*, vol. 2021, no. 1, p. 2444363, 2021.
- [7] N. K. Sekar, V. Malaghan, and D. S. Pawar, “Micro-simulation insights into the safety and operational benefits of autonomous vehicles,” *Journal of Intelligent and Connected Vehicles*, vol. 6, no. 4, pp. 202–210, 2023.
- [8] C. Hickert, S. Li, and C. Wu, “Cooperation for scalable supervision of autonomy in mixed traffic,” *IEEE Transactions on Robotics*, vol. 39, no. 4, pp. 2751–2769, 2023.
- [9] C. Hickert, Z. Yan, and C. Wu, “A data-informed analysis of scalable supervision for safety in autonomous vehicle fleets,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 986–993.
- [10] S. Mohammadian, Z. Zheng, M. M. Haque, and A. Bhaskar, “Performance of continuum models for realworld traffic flows: Comprehensive benchmarking,” *Transportation Research Part B: Methodological*, vol. 147, pp. 132–167, 2021.
- [11] V. Alexiadis, R. VanGorder, G. Cordahi, J. Sloboden *et al.*, “Guidance on the level of effort required to conduct traffic analysis using microsimulation,” United States. Department of Transportation. Federal Highway Administration . . . , Tech. Rep., 2014.
- [12] K. Boriboonsomsin and M. Barth, “Impacts of freeway high-occupancy vehicle lane configuration on vehicle emissions,” *Transportation Research Part D: Transport and Environment*, vol. 13, no. 2, pp. 112–125, 2008.
- [13] R. G. Khavas, B. Hellinga, and A. Zarinbal Masouleh, “Identifying parameters for microsimulation modeling of traffic in inclement weather,” *Transportation Research Record*, vol. 2613, no. 1, pp. 52–60, 2017.
- [14] W. Ren, X. Zhao, H. Li, and Q. Fu, “Traffic flow impact of mixed heterogeneous platoons on highways: An approach combining driving simulation and microscopic traffic simulation,” *Physica A: Statistical Mechanics and its Applications*, vol. 643, p. 129803, 2024.
- [15] A. Qu and C. Wu, “Revisiting the correlation between simulated and field-observed conflicts using large-scale traffic reconstruction,” *Accident Analysis & Prevention*, vol. 210, p. 107808, 2025.
- [16] V. Punzo and M. Montanino, “Speed or spacing? cumulative variables, and convolution of model errors and time in traffic flow models validation and calibration,” *Transportation Research Part B: Methodological*, vol. 91, pp. 21–33, 2016.
- [17] M. Treiber and A. Kesting, “Microscopic calibration and validation of car-following models—a systematic approach,” *Procedia-Social and Behavioral Sciences*, vol. 80, pp. 922–939, 2013.
- [18] A. Sharma, Z. Zheng, and A. Bhaskar, “Is more always better? the impact of vehicular trajectory completeness on car-following model calibration and validation,” *Transportation research part B: methodological*, vol. 120, pp. 49–75, 2019.

TABLE II: Parameters and Default Calibration Ranges

Category	Description	Range
Origin-Destination (OD)	Flow volumes between all feasible edge-level origin-destination pairs	[0, 15,000] veh/hr
Car-Following (CFM)	Desired minimum time headway behind a leading vehicle (τ)	[0.2, 3] s
	Maximum deceleration to avoid an accident	[6, 12] m/s ²
	Maximum acceleration the driver applies	[0.5, 5] m/s ²
	Maximum comfortable deceleration the driver applies	[1, 6] m/s ²
	Minimum gap maintained at standstill (bumper-to-bumper)	[0.2, 5] m
	Acceleration exponent (δ) controlling smoothness	[2, 6]
	Duration between vehicle decisions for speed and lane-changing	[0.9, 1.1] s
Lane-Changing (LC)	Factor scaling a vehicle's desired maximum speed	[0.9, 1.1]
	Eagerness for long-term, route-driven lane changes	[0, 1]
	Willingness to adjust speed to facilitate others' lane changes	[0, 1]
	Willingness to accept smaller gaps during lane changes	[0, 1]
	Eagerness for lane changes to gain speed	[0, 1]
Heterogeneity (HET)	Tendency to keep to the rightmost available lane	[0, 1]
	Variance control for driver behavior attributes	[0.02, 1]
Simulation (SIM)	Variance in speed factor among different drivers	[0.02, 0.05]
	Number of 'chunks' the simulation is divided into	{1, 2, 4, 6}
	Warm-up period at simulation start for network to reach stable traffic	[0, 300] s
	Duration of a single simulation step	[0.1, 1.0] s

- [19] S.-J. Kim, W. Kim, and L. R. Rilett, "Calibration of microsimulation models using nonparametric statistical techniques," *Transportation Research Record*, vol. 1935, no. 1, pp. 111–119, 2005.
- [20] Y. Hollander and R. Liu, "The principles of calibrating traffic microsimulation models," *Transportation*, vol. 35, pp. 347–362, 2008.
- [21] M. Jha, G. Gopalan, A. Garms, B. P. Mahanti, T. Toledo, and M. E. Ben-Akiva, "Development and calibration of a large-scale microscopic traffic simulation model," *Transportation Research Record*, vol. 1876, no. 1, pp. 121–131, 2004.
- [22] M. Montanino and V. Punzo, "Trajectory data reconstruction and simulation-based validation against macroscopic traffic patterns," *Transportation Research Part B: Methodological*, vol. 80, pp. 82–106, 2015.
- [23] C. Antoniou, J. Barceló, M. Breen, M. Bulles, J. Casas, E. Cipriani, B. Ciuffo, T. Djukic, S. Hoogendoorn, V. Marzano *et al.*, "Towards a generic benchmarking platform for origin-destination flows estimation/updating algorithms: Design, demonstration and validation," *Transportation Research Part C: Emerging Technologies*, vol. 66, pp. 79–98, 2016.
- [24] C. Rong, J. Ding, Y. Liu, and Y. Li, "A large-scale benchmark dataset for commuting origin-destination matrix generation," *arXiv preprint arXiv:2407.15823*, 2024.
- [25] V. Punzo, Z. Zheng, and M. Montanino, "About calibration of car-following dynamics of automated and human-driven vehicles: Methodology, guidelines and codes," *Transportation Research Part C: Emerging Technologies*, vol. 128, p. 103165, 2021.
- [26] N. Montali, J. Lambert, P. Mougin, A. Kuefler, N. Rhinehart, M. Li, C. Gulino, T. Emrich, Z. Yang, S. Whiteson *et al.*, "The waymo open sim agents challenge," *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [27] N. Geroliminis and C. F. Daganzo, "Existence of urban-scale macroscopic fundamental diagrams: Some experimental findings," *Transportation Research Part B: Methodological*, vol. 42, no. 9, pp. 759–770, 2008.
- [28] D. S. Simbeye, "Deployment of inductive loop vehicle traffic counters along trunk roads in tanzania," *Tanzania Journal of Engineering and Technology*, vol. 41, no. 04, pp. 119–132, 2022.
- [29] F. Soriguera and F. Robusté, "Estimation of traffic stream space mean speed from time aggregations of double loop detector data," *Transportation research part C: emerging technologies*, vol. 19, no. 1, pp. 115–129, 2011.
- [30] J. R. Meijer, M. A. Huijbregts, K. C. Schotten, and A. M. Schipper, "Global patterns of current and future road infrastructure," *Environmental Research Letters*, vol. 13, no. 6, p. 064006, 2018.
- [31] N. H. T. S. Administration *et al.*, "Early estimates of motor vehicle traffic fatalities and fatality rate by sub-categories in 2021," *Crash• Stats Brief Statistical Summary. Report No. DOT HS*, vol. 813, p. 118, 2021.
- [32] P. Varaiya, "Freeway performance measurement system (pems) version 4," 2004.
- [33] D. Gloudemans, Y. Wang, J. Ji, G. Zachar, W. Barbour, E. Hall, M. Cebelak, L. Smith, and D. B. Work, "I-24 motion: An instrument for freeway traffic science," *Transportation Research Part C: Emerging Technologies*, vol. 155, p. 104311, 2023.
- [34] A. Reuel-Lamparth, A. Hardy, C. Smith, M. Lamparth, M. Hardy, and M. J. Kochenderfer, "Betterbench: Assessing ai benchmarks, uncovering issues, and establishing best practices," *Advances in Neural Information Processing Systems*, vol. 37, pp. 21 763–21 813, 2024.
- [35] B. Coifman and L. Li, "A critical evaluation of the next generation simulation (ngsim) vehicle trajectory dataset," *Transportation Research Part B: Methodological*, vol. 105, pp. 362–377, 2017.
- [36] D. Gloudemans, G. Zachár, Y. Wang, J. Ji, M. Nice, M. Bunting, W. W. Barbour, J. Sprinkle, B. Piccoli, M. L. D. Monache *et al.*, "So you think you can track?" in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2024, pp. 4528–4538.
- [37] C. Zhang and L. Sun, "Bayesian calibration of the intelligent driver model," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 9308–9320, 2024.
- [38] M. Samaei, M. Ameli, J. F. Davis, S. T. McQuade, J. Lee, B. Piccoli, and A. Bayen, "Integrating multi-source data for bi-level traffic simulator calibration: A literature review and highway case study," *Available at SSRN 4775835*, 2024.
- [39] Y. Wang, F. de Souza, Y. Zhang, and D. Karbowski, "Calibrating microscopic traffic models with macroscopic data," 2024.
- [40] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2019, pp. 2623–2631.
- [41] G. Amirjamshidi and M. J. Roorda, "Multi-objective calibration of traffic microsimulation models," *Transportation letters*, vol. 11, no. 6, pp. 311–319, 2019.
- [42] J. Erdmann, "Sumo's lane-changing model," in *Modeling Mobility with Open Data: 2nd SUMO Conference 2014 Berlin, Germany, May 15-16, 2014*. Springer, 2015, pp. 105–123.